Adham Ashraf Saeed

Assignment 2 (Mapping & Normalization)

Problem 1

- DreamHome Company is designed to connect individuals interested in renting properties with their owners. It's important to note that the term 'Rent' refers to the value assigned to the property, determined through negotiation between the customer and the owner.
- To achieve the (3NF) using the provided data begin by detailing the steps for your answer. The primary key should be the Customer number. Exclude the Page number and Date from the bill shown below.

Data

Customer_Rental Table

Customer_No	*CName*	*Property_No*	*PAddress*	*RentStart*	*RentFinish*	*Rent*	*Owner_No*	*OName*
CR76	John Kay	PG4	6 Lawrence St, Glasgow	1-Jul-94	31-Aug-96	350	CO40	Tina Murphy
		PG16	5 Novar Dr, Glasgow	1-Sep-96	1-Sep-98	450	CO93	Tony Shaw
CR56	Aline Stewart	PG4	6 Lawrence St, Glasgow	1-Sep-92	10-June-94	350	CO40	Tina Murphy
		PG36	2 Manor Rd, Glasgow	10-Oct-94	1-Dec-95	375	CO93	Tony Shaw
		PG16	5 Novar Dr, Glasgow	1-Jan-96	10-Aug-96	450	CO93	Tony Shaw

Customer_No	CName	Property_No	PAddress	RentStart	RentFinish	Rent	Owner_No	OName
CR76	John Kay	PG4	6 Lawrence St, Glasgow	1-Jul-94	31-Aug-96	350	CO40	Tina Murphy
		PG16	5 Novar Dr, Glasgow	1-Sep-96	1-Sep-98	450	CO93	Tony Shaw
CR56	Aline Stewart	PG4	6 Lawrence St, Glasgow	1-Sep-92	10-Jun-94	350	CO40	Tina Murphy
		PG36	2 Manor Rd, Glasgow	10-Oct-94	1-Dec-95	375	CO93	Tony Shaw
		PG16	5 Novar Dr, Glasgow	1-Jan-96	10-Aug-96	450	CO93	Tony Shaw

1NF

Customer_No	CName	Property_No	PAddress	RentStart	RentFinish	Rent	Owner_No	OName
CR76	John Kay	PG4	6 Lawrence St, Glasgow	1-Jul-94	31-Aug-96	350	CO40	Tina Murphy
CR76	John Kay	PG16	5 Novar Dr, Glasgow	1-Sep-96	1-Sep-98	450	CO93	Tony Shaw
CR56	Aline Stewart	PG4	6 Lawrence St, Glasgow	1-Sep-92	10-Jun-94	350	CO40	Tina Murphy
CR56	Aline Stewart	PG36	2 Manor Rd, Glasgow	10-Oct-94	1-Dec-95	375	CO93	Tony Shaw
CR56	Aline Stewart	PG16	5 Novar Dr, Glasgow	1-Jan-96	10-Aug-96	450	CO93	Tony Shaw

2NF

Customer	
Customer_No (PK)	CName
CR76	John Kay
CR56	Aline Stewart

Property			
Property_No (PK)	PAddress	Owner_No	OName
PG4	6 Lawrence St, Glasgow	CO40	Tina Murphy
PG16	5 Novar Dr, Glasgow	CO93	Tony Shaw
PG36	2 Manor Rd, Glasgow	CO93	Tony Shaw

Rental				
Customer_No (PK)	Property_No (PK)	RentStart	RentFinish	Rent
CR76	PG4	1-Jul-94	31-Aug-96	$350
CR76	PG16	1-Sep-96	1-Sep-98	$450
CR56	PG4	1-Sep-92	10-Jun-94	$350
CR56	PG36	10-Oct-94	1-Dec-95	$375
CR56	PG16	1-Jan-96	10-Aug-96	$450

3NF

Customer

Customer_No (PK)	CName
CR76	John Kay
CR56	Aline Stewart

Property

Property_No (PK)	PAddress	Owner_No
PG4	6 Lawrence St, Glasgow	CO40
PG16	5 Novar Dr, Glasgow	CO93
PG36	2 Manor Rd, Glasgow	CO93

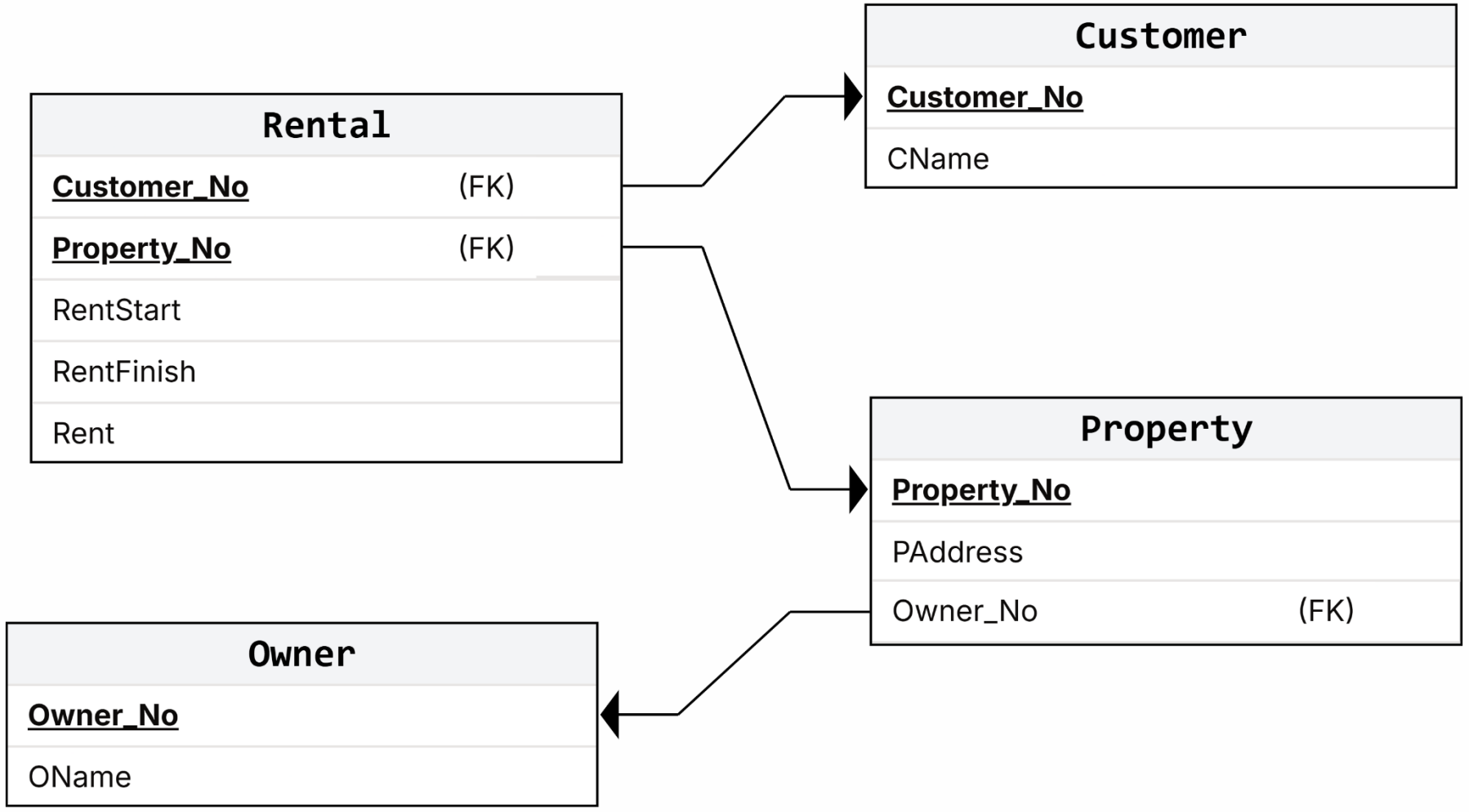
Rental

Customer_No (PK)	Property_No (PK)	RentStart	RentFinish	Rent
CR76	PG4	1-Jul-94	31-Aug-96	$350
CR76	PG16	1-Sep-96	1-Sep-98	$450
CR56	PG4	1-Sep-92	10-Jun-94	$350
CR56	PG36	10-Oct-94	1-Dec-95	$375
CR56	PG16	1-Jan-96	10-Aug-96	$450

Owner

Owner_No (PK)	OName
CO40	Tina Murphy
CO93	Tony Shaw

Mapping


Customer
Customer_No
CName
Rental
Customer_No (FK)
Property_No (FK)
RentStart
RentFinish
Rent
Property
Property_No
PAddress
Owner_No (FK)
Owner
Owner_No
OName

Problem 2

- The provided report contains detailed information concerning the organization's projects and the employees assigned to them. The records illustrate a many-to-many relationship: a project can involve multiple employees, and an employee can work on multiple projects. Additionally, each job classification is associated with a specific hourly charge rate (CHG/Hour). Your task is to apply the First, Second, and Third Normal Forms (1NF, 2NF, and 3NF) to this data.

Data

PROJ_NUM	PROJECT_NAME	EMP_NUM	EMPLOYEE_NAME	JOB_CLASS	CHG_HOUR	HOURS_BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	$85.50	23.8	$2,034.90
		101	John G. News	Database Designer	$105.00	19.4	$2,037.00
		105	Alice K. Johnson	Database Designer	$105.00	35.7	$3,748.50
		106	William Smithfield	Programmer	$35.75	12.6	$450.45
		102	David H. Senior	Systems Analyst	$96.75	23.8	$2,302.65
18	Amber Wave	114	Annelise Jones	Applications Designer	$48.10	25.6	$1,231.36
		118	James J. Frommer	General Support	$18.36	45.3	$831.71
		104	Anne K. Ramoras	Systems Analyst	$96.75	32.4	$3,134.70
		112	Darlene M. Smithson	DSS Analyst	$45.95	45.0	$2,067.75
22	Rolling Tide	105	Alice K. Johnson	Database Designer	$105.00	65.7	$6,898.50
		104	Anne K. Ramoras	Systems Analyst	$96.75	48.4	$4,682.70
		113	Delbert K. Joenbrood	Applications Designer	$48.10	23.6	$1,135.16
		111	Geoff B. Wabash	Clerical Support	$26.87	22.0	$591.14
		106	William Smithfield	Programmer	$35.75	12.8	$457.60
25	Starflight	107	Maria D. Alonzo	Programmer	$35.75	25.6	$915.20
		115	Travis B. Bawangi	Systems Analyst	$96.75	45.8	$4,431.15
		101	John G. News	Database Designer	$105.00	56.3	$5,911.50
		114	Annelise Jones	Applications Designer	$48.10	33.1	$1,592.11
		108	Ralph B. Washington	Systems Analyst	$96.75	23.6	$2,283.30
		118	James J. Frommer	General Support	$18.36	30.5	$559.98
		112	Darlene M. Smithson	DSS Analyst	$45.95	41.4	$1,902.33

TABLE 5.1 A Sample Report Layout

PROJ. NUM.	PROJECT NAME	EMPLOYEE NUMBER	EMPLOYEE NAME	JOB CLASS	CHG/ HOUR	HOURS BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	$ 85.50	23.8	$ 2,011.10
		101	John G. News	Database Designer	$105.00	19.4	$ 2,037.00
		105	Alice K. Johnson*	Database Designer	$105.00	35.7	$ 3,748.50
		106	William Smithfield	Programmer	$ 35.75	12.6	$ 450.45
		102	David H. Senior	Systems Analyst	$ 96.75	23.8	$ 2,302.65
				Subtotal			**$10,549.70**
18	Amber Wave	114	Annelise Jones	Applications Designer	$ 48.10	25.6	$ 1,183.26
		118	James J. Frommer	General Support	$ 18.36	45.3	$ 831.71
		104	Anne K. Ramoras*	Systems Analyst	$ 96.75	32.4	$ 3,135.70
		112	Darlene M. Smithson	DSS Analyst	$ 45.95	45.0	$ 2,021.80
				Subtotal			**$ 7,172.47**
22	Rolling Tide	105	Alice K. Johnson	Database Designer	$105.00	65.7	$ 6,793.50
		104	Anne K. Ramoras	Systems Analyst	$ 96.75	48.4	$ 4,682.70
		113	Delbert K. Joenbrood*	Applications Designer	$ 48.10	23.6	$ 1,135.16
		111	Geoff B. Wabash	Clerical Support	$ 26.87	22.0	$ 591.14
		106	William Smithfield	Programmer	$ 35.75	12.8	$ 457.60
				Subtotal			**$13,660.10**
25	Starflight	107	Maria D. Alonzo	Programmer	$ 35.75	25.6	$ 879.45
		115	Travis B. Bawangi	Systems Analyst	$ 96.75	45.8	$ 4,431.15
		101	John G. News*	Database Designer	$105.00	56.3	$ 5,911.50
		114	Annelise Jones	Applications Designer	$ 48.10	33.1	$ 1,592.11
		108	Ralph B. Washington	Systems Analyst	$ 96.75	23.6	$ 2,283.30
		118	James J. Frommer	General Support	$ 18.36	30.5	$ 559.98
		112	Darlene M. Smithson	DSS Analyst	$ 45.95	41.4	$ 1,902.33
				Subtotal			**$17,559.02**
				Total			**$48,942.09**

Note: * indicates project leader.

1NF

PROJ_NUM	PROJECT_NAME	EMP_NUM	EMPLOYEE_NAME	JOB_CLASS	CHG_HOUR	HOURS_BILLED	TOTAL_CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	$85.50	23.8	$2,034.90
15	Evergreen	101	John G. News	Database Designer	$105.00	19.4	$2,037.00
15	Evergreen	105	Alice K. Johnson	Database Designer	$105.00	35.7	$3,748.50
15	Evergreen	106	William Smithfield	Programmer	$35.75	12.6	$450.45
15	Evergreen	102	David H. Senior	Systems Analyst	$96.75	23.8	$2,302.65
18	Amber Wave	114	Annelise Jones	Applications Designer	$48.10	25.6	$1,231.36
18	Amber Wave	118	James J. Frommer	General Support	$18.36	45.3	$831.71
18	Amber Wave	104	Anne K. Ramoras	Systems Analyst	$96.75	32.4	$3,134.70
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	$45.95	45.0	$2,067.75
22	Rolling Tide	105	Alice K. Johnson	Database Designer	$105.00	65.7	$6,898.50
22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	$96.75	48.4	$4,682.70
22	Rolling Tide	113	Delbert K. Joenbrood	Applications Designer	$48.10	23.6	$1,135.16
22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	$26.87	22.0	$591.14
22	Rolling Tide	106	William Smithfield	Programmer	$35.75	12.8	$457.60
25	Starflight	107	Maria D. Alonzo	Programmer	$35.75	25.6	$915.20
25	Starflight	115	Travis B. Bawangi	Systems Analyst	$96.75	45.8	$4,431.15
25	Starflight	101	John G. News	Database Designer	$105.00	56.3	$5,911.50
25	Starflight	114	Annelise Jones	Applications Designer	$48.10	33.1	$1,592.11
25	Starflight	108	Ralph B. Washington	Systems Analyst	$96.75	23.6	$2,283.30
25	Starflight	118	James J. Frommer	General Support	$18.36	30.5	$559.98
25	Starflight	112	Darlene M. Smithson	DSS Analyst	$45.95	41.4	$1,902.33

2NF

Project

PROJ_NUM (PK)	PROJECT_NAME
15	Evergreen
18	Amber Wave
22	Rolling Tide
25	Starflight

Employee

EMP_NUM (PK)	EMPLOYEE_NAME	JOB_CLASS	CHG_HOUR
103	June E. Arbough	Elec. Engineer	$85.50
101	John G. News	Database Designer	$105.00
105	Alice K. Johnson	Database Designer	$105.00
106	William Smithfield	Programmer	$35.75
102	David H. Senior	Systems Analyst	$96.75
114	Annelise Jones	Applications Designer	$48.10
118	James J. Frommer	General Support	$18.36
104	Anne K. Ramoras	Systems Analyst	$96.75
112	Darlene M. Smithson	DSS Analyst	$45.95
113	Delbert K. Joenbrood	Applications Designer	$48.10
111	Geoff B. Wabash	Clerical Support	$26.87
107	Maria D. Alonzo	Programmer	$35.75
115	Travis B. Bawangi	Systems Analyst	$96.75
108	Ralph B. Washington	Systems Analyst	$96.75

Assignment

PROJ_NUM (PK)	EMP_NUM (PK)	HOURS_BILLED	TOTAL_CHARGE
15	103	23.8	$2,034.90
15	101	19.4	$2,037.00
15	105	35.7	$3,748.50
15	106	12.6	$450.45
15	102	23.8	$2,302.65
18	114	25.6	$1,231.36
18	118	45.3	$831.71
18	104	32.4	$3,134.70
18	112	45.0	$2,067.75
22	105	65.7	$6,898.50
22	104	48.4	$4,682.70
22	113	23.6	$1,135.16
22	111	22.0	$591.14
22	106	12.8	$457.60
25	107	25.6	$915.20
25	115	45.8	$4,431.15
25	101	56.3	$5,911.50
25	114	33.1	$1,592.11
25	108	23.6	$2,283.30
25	118	30.5	$559.98
25	112	41.4	$1,902.33

3NF

Project

PROJ_NUM (PK)	PROJECT_NAME
15	Evergreen
18	Amber Wave
22	Rolling Tide
25	Starflight

Employee

EMP_NUM (PK)	EMPLOYEE_NAME	JOB_CLASS
103	June E. Arbough	Elec. Engineer
101	John G. News	Database Designer
105	Alice K. Johnson	Database Designer
106	William Smithfield	Programmer
102	David H. Senior	Systems Analyst
114	Annelise Jones	Applications Designer
118	James J. Frommer	General Support
104	Anne K. Ramoras	Systems Analyst
112	Darlene M. Smithson	DSS Analyst
113	Delbert K. Joenbrood	Applications Designer
111	Geoff B. Wabash	Clerical Support
107	Maria D. Alonzo	Programmer
115	Travis B. Bawangi	Systems Analyst
108	Ralph B. Washington	Systems Analyst

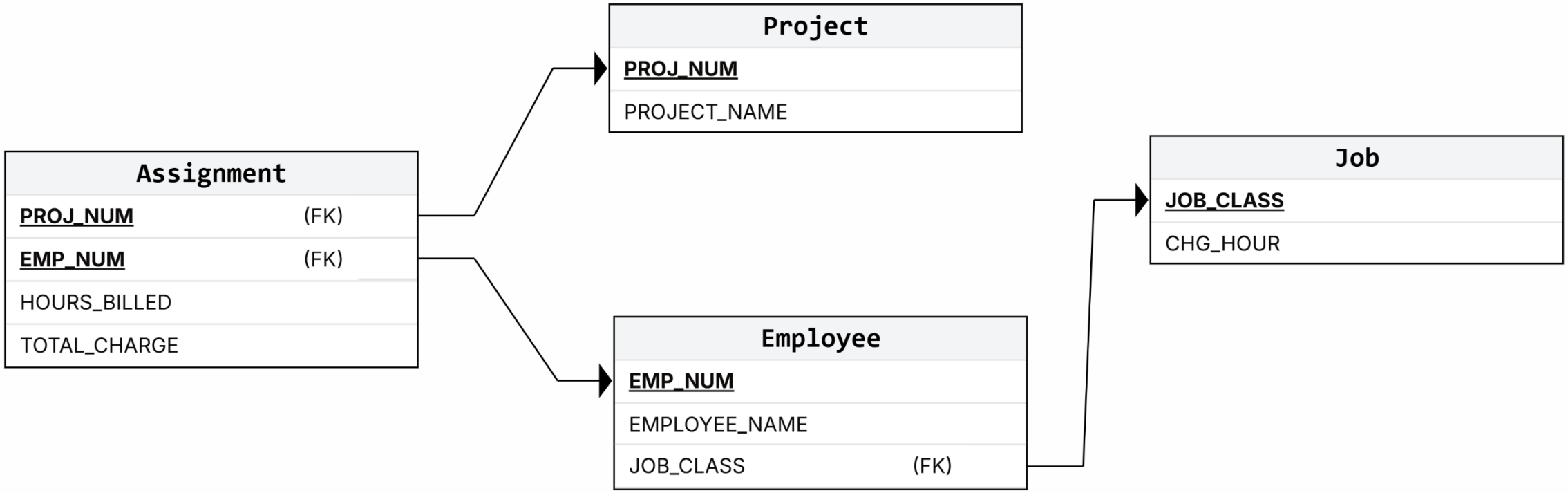
Job

JOB_CLASS(PK)	CHG_HOUR
Elec. Engineer	$85.50
Database Designer	$105.00
Programmer	$35.75
Systems Analyst	$96.75
Applications Designer	$48.10
General Support	$18.36
DSS Analyst	$45.95
Clerical Support	$26.87

Assignment

PROJ_NUM (PK)	EMP_NUM (PK)	HOURS_BILLED	TOTAL_CHARGE
15	103	23.8	$2,034.90
15	101	19.4	$2,037.00
15	105	35.7	$3,748.50
15	106	12.6	$450.45
15	102	23.8	$2,302.65
18	114	25.6	$1,231.36
18	118	45.3	$831.71
18	104	32.4	$3,134.70
18	112	45.0	$2,067.75
22	105	65.7	$6,898.50
22	104	48.4	$4,682.70
22	113	23.6	$1,135.16
22	111	22.0	$591.14
22	106	12.8	$457.60
25	107	25.6	$915.20
25	115	45.8	$4,431.15
25	101	56.3	$5,911.50
25	114	33.1	$1,592.11
25	108	23.6	$2,283.30
25	118	30.5	$559.98
25	112	41.4	$1,902.33

Mapping


Project
PROJ_NUM
PROJECT_NAME
Assignment
PROJ_NUM (FK)
EMP_NUM (FK)
HOURS_BILLED
TOTAL_CHARGE
Job
JOB_CLASS
CHG_HOUR
Employee
EMP_NUM
EMPLOYEE_NAME
JOB_CLASS (FK)